An un-null net with no un-null sequence of its terms

A counterexample to Question 2.14 of arXiv:1605.03538

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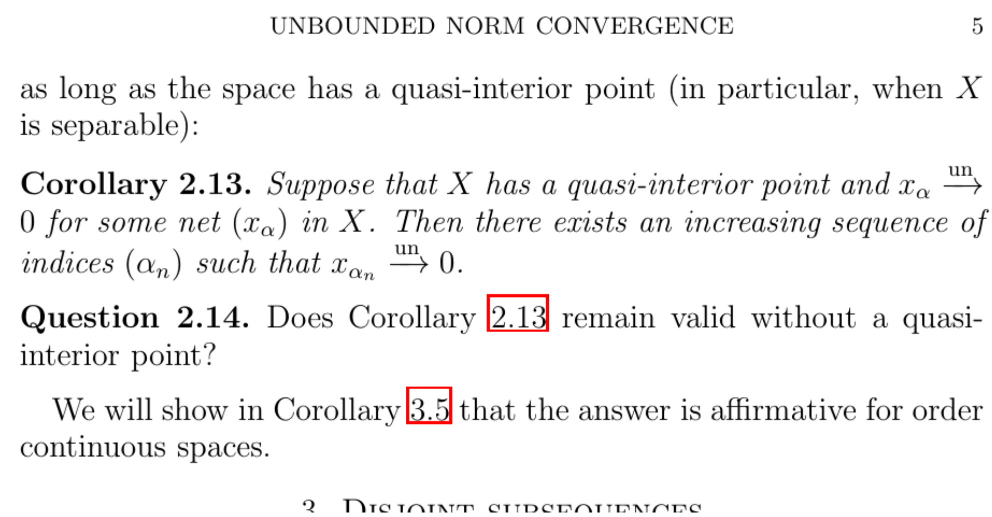
Status. Candidate counterexample, likely valid, pending human review. Question 2.14 has a negative answer. In fact, there is a positive normalized un-null net from which *no* sequence of terms is un-null, even if the indices need not be increasing.

1 Source question

For a Banach lattice X, a net (x_α) is *un-null* if

$$\big\| |x_\alpha| \wedge u \big\| \longrightarrow 0 \qquad (u \in X_+).$$

Corollary 2.13 of the source proves that, when X has a quasi-interior point, every un-null net contains an increasing sequence of indices whose terms are still un-null. Question 2.14 on source PDF page 5 asks whether this remains true without a quasi-interior point [1].



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as long as the space has a quasi-interior point (in particular, when X is separable):

Corollary 2.13. *Suppose that X has a quasi-interior point and $x_\alpha \xrightarrow{\text{un}} 0$ for some net (x_α) in X. Then there exists an increasing sequence of indices (α_n) such that $x_{\alpha_n} \xrightarrow{\text{un}} 0$.*

Question 2.14. Does Corollary 2.13 remain valid without a quasi-interior point?

We will show in Corollary 3.5 that the answer is affirmative for order continuous spaces.

2 Proof intuition

The failure comes from a clash between two countabilities. Every element of $\ell_\infty^c(\omega_1)$ sees only countably many coordinates, so a net can eventually escape the support of each fixed test vector. But after anyone chooses countably many terms of the net, all coordinates used by those terms again form one countable set. Its indicator is a single legal test vector which detects every chosen term with norm one.

3 The counterexample

Let $\Gamma = \omega_1$, the first uncountable ordinal, and let

$$X = \ell_\infty^c(\Gamma) := \{x \in \ell_\infty(\Gamma) : \operatorname{supp} x \text{ is countable}\}$$

with the coordinatewise order and the supremum norm. Let $\mathcal{A} = [\Gamma]^{\le \aleph_0}$ be the directed set of countable subsets of Γ, ordered by inclusion. For $C \in \mathcal{A}$, put

$$\gamma_C = \min(\Gamma \setminus C), \qquad x_C = e_{\gamma_C}.$$

The minimum exists because C is countable whereas Γ is not.

Theorem 1. *The space X is a Banach lattice with no quasi-interior point, and the net $(x_C)_{C\in\mathcal{A}}$ is positive, normalized, and un-null. Nevertheless, for every sequence (C_n) in $\mathcal{A}$ there is a single $u \in X_+$ such that*

$$\big\| \, |x_{C_n}| \wedge u \, \big\| = 1 \qquad (n \in \mathbb{N}).$$

Consequently, no sequence of terms of this net is un-null; in particular, Question 2.14 has a negative answer.

Proof. The space X is a vector sublattice and an ideal of $\ell_\infty(\Gamma)$. It is norm closed: if $z_k \in X$ and $z_k \to z$ uniformly, then

$$\operatorname{supp} z \subseteq \bigcup_{k=1}^{\infty} \operatorname{supp} z_k,$$

which is countable. Thus X is a Banach lattice.

No $e \in X_+$ is quasi-interior. Indeed, $\operatorname{supp} e$ is countable, so one may choose $\eta \in \Gamma \setminus \operatorname{supp} e$. Every vector in the ideal generated by e, and hence every vector in its norm closure, vanishes at η; the vector e_η does not. Thus the closed ideal generated by e is proper.

Now fix $u \in X_+$ and put $D = \operatorname{supp} u$. This set is countable and hence is an index in $\mathcal{A}$. Whenever $C \ge D$, the definition gives $\gamma_C \notin C \supseteq D$. Therefore

$$|x_C| \wedge u = e_{\gamma_C} \wedge u = 0.$$

This holds eventually for every $u \in X_+$, so $x_C \to 0$ in the un-topology. Also $x_C \ge 0$ and $\|x_C\|_\infty = 1$ for every C.

Finally, let (C_n) be any sequence of indices and set

$$E = \{\gamma_{C_n} : n \in \mathbb{N}\}, \qquad u = \mathbf{1}_E.$$

The set E is countable, so $u \in X_+$. Since $u(\gamma_{C_n}) = 1$, we have

$$|x_{C_n}| \wedge u = e_{\gamma_{C_n}} \quad \text{and hence} \quad \big\| \, |x_{C_n}| \wedge u \, \big\|_\infty = 1$$

for every n. Thus (x_{C_n}) is not un-null. The argument does not use that (C_n) is increasing, proving the stronger assertion. $\square$

4 Strength, checks, and literature boundary

The initial search through c_0-sums failed because their uniformly small tails permit a diagonal sequence extraction. Replacing c_0 support by arbitrary countable support is the decisive upgrade: fixed un-tests remain countable, while the indicator of all coordinates used by a proposed sequence remains in the lattice. The example is therefore stronger than required: the net consists of positive norm-one atoms, and no sequence of its values is un-null. The finite incidence identities underlying the two support arguments are independently exercised in `code/verify_incidence.py`; the infinite conclusion itself is the set-theoretic proof above.

A bounded primary-source search used the exact question, un-null nets, embedded sequences, the un-topology, and countably supported bounded functions. The 2023 paper of Deng and de Jeu proves that every un-convergent net has an embedded sequence that is uo-convergent, and obtains a sequence that is also un-convergent under order continuity [2]; it does not state the unrestricted conclusion refuted here. No primary arXiv source stating this counterexample was found. Correctness confidence is high; novelty confidence is moderate pending a broader expert search.

The source contains a second open question: whether weak nullity of every norm-bounded positive un-null net forces X^* to be order continuous. That question was subsequently answered affirmatively by Elbour's Theorem 2.16 [3]. Indeed, the positive-net hypothesis applied to $|x_\alpha|$ makes every functional vanish on every norm-bounded un-null net, and the cited theorem with scalar target yields order continuity of X^*. This literature fact is separate from Theorem 1.

References

[1] Y. Deng, M. O'Brien, and V. G. Troitsky, *Unbounded norm convergence in Banach lattices*, arXiv:1605.03538.

[2] Y. Deng and M. de Jeu, *Embedded unbounded order convergent sequences in topologically convergent nets in vector lattices*, arXiv:2309.03027.

[3] A. Elbour, *Some properties on the unbounded absolute weak convergence in Banach lattices*, arXiv:2004.10691.